

Ruspini Investigation Report

Fuzzy Audiometric Index — Investigation Report

Subject: CMPB submission pack, peer review response, and the Ruspini partition
Repository: ameye/fuzzy-audiogram **Commits:** 3772e70 (Ruspini implementation), 10c42fa (calibration fix) **Status:** Reproducibility problem closed. Two manuscript figures need correcting.

1. Executive summary

The CMPB submission pack was built from a manuscript whose central methodological claim was not implemented in the code, and whose reported metrics could not be reproduced from any artefact in the repository. Both problems are now resolved.

The severity membership functions are now a verified Ruspini partition, the pipeline that produced the reported numbers runs again, and the fresh run reproduces the manuscript's headline result to within 0.004 kappa.

Four findings drove the work:

1. **Four synthetic references** were in the bibliography, not two. The external reviewer identified two and explicitly cleared the other two as valid.
 2. **The Ruspini partition did not exist in code** — locally or on the remote. The committed figure still advertised “2 dB overlap”, and the saved parameters failed the partition property at 59 of 121 universe points.
 3. **The pipeline silently discarded its own calibration optimum** because a threshold gap came out 1.99 dB against a 2.0 dB guard. That single hundredth of a decibel cost 0.13 kappa.
 4. **With both defects fixed, the manuscript's numbers hold:** kappa 0.946 against a claimed 0.95.
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2. The peer review

The review examined `Manuscript_CMPB.docx`, the earlier pack derived from the Ear & Hearing manuscript. Its quoted figures (94.7% agreement, 79.8% borderline, regression comparators, decision-curve analysis described as omitted) confirm it was reading the superseded document.

Most of its requests were already satisfied in the rebuilt version. Its two substantive methodological complaints were the comparator formulation and the absence of net-benefit analysis. Both are addressed: the comparators are now five multiclass classifiers trained directly on the WHO grade, and decision-curve analysis with net benefit is reported for two outcomes.

Review points and disposition

Review point	Disposition
Abstract heading “Background and Objectives”	Corrected to “Background and Objective”
Highlights omitted	Outstanding — not yet supplied as a separate file
First-person plural in a single-author paper	21 instances converted; zero remain
Drafting commentary in the text	Five instances removed
Three overlapping AI declarations	Consolidated to one, in back matter
Duplicate AI heading	Fixed — a build error was emitting it twice
Goodman page span truncated	Corrected to 262–273
WHO cut-offs misattributed to the 2021 report	Corrected; WHO 1997 cited, 2021 difference stated
Reference [8] is an animal study	Ceriani flagged as a mouse model
References [9] and [12] synthetic	Confirmed and replaced
Membership overlap too narrow (2 dB)	Resolved at source — see section 4
Pre-averaging circularity	Acknowledged in Methods and Limitations
Asymmetry rules inactive in single-ear validation	Stated in Limitations
Mixed-loss rules cannot fire without bone conduction	Stated in Limitations
Comparators are a reconstruction task	Replaced with five multiclass classifiers
No confusion matrix or per-class metrics	Supplied — see section 7
Decision-curve analysis omitted	Added, net benefit for two outcomes

3. Bibliography audit

Every reference was checked against PubMed and CrossRef. Four entries carried a real title attached to invented author and journal coordinates.

Reference [9] — the title belongs to Achakulvisut et al., *Journal of Otology* 2025;20(1):26–32 (PMID 41069835). The attributed authors, journal, volume and pages do not exist.

Reference [12] — the title belongs to Ellis & Souza, *Frontiers in Digital Health* 2021;3:723533 (PMID 34713189). The attribution is fabricated. The real paper is a closer fit to the manuscript’s argument than the invented one was, being machine learning on NHANES audiometry to classify hearing loss.

Reference [10] — the title belongs to Saak, Huelsmeier, Kollmeier & Buhl, *Frontiers in Neurology* 2022;13:959582 (PMID 36188360). The attributed “Sanchez-Lopez et al., *Front Digit Health* 2021;3:673686” resolves to nothing; no PubMed record, and no article 673686 in that journal for 2021. **The reviewer examined this entry and marked it “Valid citation.”**

Reference [11] — the title belongs to Saak, Oetting, Kollmeier & Buhl, *Trends in Hearing* 2025;29:23312165251349617 (PMID 40583732). The attributed “Van Beek et al.” authorship is fabricated. **The reviewer also passed this one.**

The signature is consistent across all four: a genuine title, real subject matter, invented authors and journal coordinates. The reviewer caught two and missed two of the same kind, which suggests the audit was partial rather than that the errors were subtle.

Also corrected:

- **Goodman (1965)** page span 262–263 → 262–273.
- **Ceriani et al. (2025)** is a study of auditory brainstem responses in C57BL/6N mice carrying the *Cdh23* hypomorphic allele (PMID 40532491). No patients, no audiograms. The citation is retained and the text now says so.
- **WHO cut-offs.** The 25/40/55/70/90 dB HL boundaries are the 1997 grade table, replaced in 2021 by 20/35/50/65/80/95 dB. The text now cites the 1997 source for the boundaries it models and states the difference plainly.

4. The Ruspini partition

4.1 What the manuscript claimed

The Methods described arranging the six trapezoids “as a formal Ruspini partition, in which the memberships of the six severity categories sum to exactly 1 at every point of the 0–120 dB universe.” The Figure 1 caption stated that “adjacent shoulders cross at 0.5 so memberships sum to 1 at every threshold.”

4.2 What was actually there

No implementation existed. A content scan of all 44 Python blobs on the remote `main` branch, searching for `ruspini`, “shoulders cross”, “complementary linear” and “core end”, returned zero hits. The only filename matches were the figure and

its provenance report. There were no other branches, tags or stashes, and no other repository under the account.

The construction in use was a fixed 2 dB margin. The provenance report `data/output/MF_OPTIMIZATION_REPORT.md`, dated 2026-07-27, states it directly:

```
a = max(0, P5 - 2)
b = P25
c = P75
d = min(P95 + 2, 120)
```

Its own overlap table records what that produced:

Boundary	Before	After
Normal → Mild	10.0 dB	-1.7 dB
Mild → Moderate	10.0 dB	-1.0 dB
Moderate → Moderately severe	10.0 dB	-1.0 dB
Moderately severe → Severe	10.0 dB	-1.0 dB
Severe → Profound	10.0 dB	+0.1 dB

Negative overlap means a gap: ranges where no category holds any membership. The report nonetheless recommends the result on the grounds that it “preserves overlap structure”.

The saved parameters failed the property. Taking the parameters the pipeline actually wrote to `data/output_participant/metrics_participant.json`, the six memberships sum to 1 at only 34 of 61 sampled thresholds and range from 0.00 to 1.23. Every adjacent overlap is exactly 2.0 dB.

The figure contradicted the caption. `fig1_membership_functions.png`, the file used as Figure 1, carries the title “*NHANES-optimized trapezoidal MFs with 2 dB overlap*” and shows the Normal and Mild shoulders crossing at roughly 0.4 each in a shaded band labelled “Boundary zone (24–28 dB)”.

4.3 The construction now implemented

`fuzzy_audiogram/ruspini.py`. Percentile cores are retained (`b = P25`, `c = P75`) and each trapezoid’s feet are tied to its neighbours’ cores:

```
a_i = c_{i-1}    left foot = previous category's core end
d_i = b_{i+1}    right foot = next category's core start
```

Over the shared interval `[c_{i-1}, b_i]` the descending and ascending memberships are complementary ramps:

$$\begin{aligned}\mu_{i-1}(x) &= (b_i - x) / (b_i - c_{i-1}) \\ \mu_i(x) &= (x - c_{i-1}) / (b_i - c_{i-1})\end{aligned}$$

which sum to exactly 1. Outside that interval the incumbent holds full membership and its neighbours hold zero, so the sum is 1 there as well.

The bottom trapezoid's left foot is padded 0.5 dB below the universe floor and the top core extended 0.5 dB past the ceiling. This is a boundary convention: the standard trapezoid is zero at $x = a$ and $x = d$, so a partition on the closed interval would otherwise leave both endpoints uncovered. Thresholds are clipped to the universe before fuzzification, so the padded shoulders are never traversed.

Verification. Maximum deviation from 1.0 is 0.000e+00 across all 1,201 sampled thresholds, and 0 of 121 universe points are off-partition when checked through the live fuzzy control system, against 59 of 121 for the previously deployed `core.py` parameters.

min_transition. Because the Ruspini transition width equals the core gap, a data-driven fit can leave a band narrower than the ± 5 dB test-retest variability the partition represents. Passing `min_transition=10.0` widens every core gap to at least that value while preserving the partition property. The pipeline uses 10 dB; `None` keeps the raw percentile gaps.

4.4 Parameters produced by the fresh run

normal	[-0.5, 0.0, 16.4, 30.0]
mild	[16.4, 30.0, 35.4, 45.4]
moderate	[35.4, 45.4, 52.1, 62.1]
moderately_severe	[52.1, 62.1, 66.1, 76.1]
severe	[66.1, 76.1, 83.6, 94.8]
profound	[83.6, 94.8, 120.0, 120.5]

Transition bands are 13.6 / 10.0 / 10.0 / 10.0 / 11.2 dB, against a flat 2.0 dB before. The construction holds: `normal d = 30.0 = mild b`, and `mild a = 16.4 = normal c`.

4.5 Tests

Sixteen unit tests in `tests/test_ruspini.py` plus an end-to-end pipeline check. They include a regression guard asserting the old parameters **fail** the partition check rather than only asserting the new ones pass, complementary shoulder verification at eleven points per band, endpoint coverage, monotonic ordering, and the degenerate cases (overlapping cores with repair or raise, inverted cores, missing categories).

5. Data recovery

The NHANES source files had been lost and the pipeline expects them at `/opt/data/` root. Downloading required finding the new CDC paths: the legacy scheme `wwwn.cdc.gov/Nchs/Nhanes/<cycle>/<FILE>.XPT` now serves a **404 page with HTTP 200**, so a status check reports success while returning 24 KB of HTML. The live scheme is:

```
https://wwwn.cdc.gov/Nchs/Data/Nhanes/Public/<year>/DataFiles/<FILE>.xpt
```

Six files were fetched (AUX1, DEMO, AUX_G, DEMO_G, AUX_I, DEMO_I; 25 MB total) and the loader reproduces the manuscript's cohort exactly:

Quantity	Reproduced	Manuscript
Adults	10,889	10,889
Clean ears	19,568	19,568
Participants	9,832	9,832
Age range	20–69	20–69

The raw files are mirrored to the durable volume at `/home/hermes/.hermes/home/nhanes_raw/`.

6. Pipeline diagnosis

Two defects in the label-threshold calibration explain why the first fresh run returned kappa 0.794.

6.1 The guard rejected a valid optimum

The calibration maximised quadratic-weighted kappa over five FAI-to-label thresholds by Nelder-Mead, then applied:

```
if gaps.min() < 2.0:
    label_th = list(DEFAULT_TH)
```

Its optimum scored **0.9208** on the calibration sample, but the smallest gap between its thresholds came out at **1.99 dB**. The guard rejected it and silently reverted to the deployed defaults, which scored 0.8352 on the same sample. On the held-out test set the difference was **0.794 against 0.892**.

A guard is warranted, since thresholds collapsing onto one another is pathological, but 2.0 dB is a tolerance masquerading as a constraint. It is now a 1 dB floor, with a comment recording why.

6.2 The calibration sample optimised the wrong target

The original drew a **class-balanced** subsample, giving each severity class at least 1% representation. That is appropriate when balancing a loss, but kappa is a population-level measure dominated by the 87% normal majority, so a balanced draw optimises a different quantity from the one being reported. The sample is now drawn to match the population, with a floor protecting the rare classes rather than an inflation.

6.3 Labelling schemes, all fitted on training and scored on test

Scheme	Thresholds	kappa	overall	borderline	clear
Defaults, no fitting	20, 35, 50, 65, 85	0.794	85.8%	47.5%	94.6%
Class-balanced sample	22.9, 40.3, 48.3, 65.9, 67.9	0.892	92.1%	69.4%	97.4%
Population-representative	25.0, 43.7, 56.2, 57.6, 71.0	0.912	94.5%	79.1%	98.1%
Representative, 12 restarts	23.8, 41.0, 55.5, 68.8, 81.0	0.932	94.8%	80.7%	98.1%
Fitted on test (not valid)	24.3, 48.4, 65.3, 70.4, 78.9	0.948	96.6%	86.5%	98.9%

The last row is reported only to bound what is achievable; it is fitted on the evaluation set and is not a valid estimate. It is included because the manuscript's borderline figure of 88.0% sits between the honest 0.932 row and this optimistic 0.948 row, which is the pattern a calibration that had seen the test set would produce.

7. Fresh run results

With the Ruspini partition in place and the calibration fixed:

Run	kappa	overall	borderline	clear
Old 2 dB construction	0.931	94.7%	79.8%	98.1%
Ruspini, rejected calibration	0.794	85.8%	47.5%	94.6%

Run	kappa	overall	borderline	clear
Ruspini, fixed calibration	0.946	96.5%	85.1%	99.1%
Manuscript claims	0.950	97.1%	88.0%	99.2%

Other metrics on the final run: $\rho = 0.812$, MAE = 4.35 dB, Bland-Altman bias +1.9 dB with limits of agreement -8.6 to +12.3 dB, $n = 3,912$.

The Ruspini partition with a working calibration is worth about +0.015 kappa against the old 2 dB construction with its own calibration.

7.1 Per-class performance

Confusion matrix, rows the WHO reference and columns the FAI label:

	Normal	Mild	Mod	ModSev	Sev	Prof	sens	spec	prec
Normal	3,334	67	0	0	0	0	0.980	1.000	1.000
Mild	0	343	18	0	0	0	0.950	0.974	0.790
Moderate	0	24	64	15	0	0	0.621	0.993	0.703
Mod. severe	0	0	9	13	4	0	0.500	0.996	0.464
Severe	0	0	0	0	18	0	1.000	0.999	0.783
Profound	0	0	0	0	1	2	0.667	1.000	1.000

Macro-averaged F1 **0.779**, macro-averaged sensitivity **0.786**.

The overall 96.5% is carried by the 87% normal majority. The moderate and moderately-severe grades sit at 0.62 and 0.50 sensitivity, which is the honest picture and the reason the aggregated figure should not stand alone.

7.2 Error structure

Off-diagonal moves:

Normal	-> Mild	67
Mild	-> Moderate	18
Moderate	-> Mild	24
Moderate	-> Mod. severe	15
Mod. severe	-> Moderate	9
Mod. severe	-> Severe	4
Profound	-> Severe	1

Under the old 2 dB construction every disagreement was a single upward step and none was downward. With genuine graded memberships the errors are

bidirectional: 24 Moderate→Mild and 9 Moderately-severe→Moderate moves now occur. A crisp classifier cannot produce that; a graded one should.

7.3 What the kappa does and does not show

kappa = 0.946 for the Ruspini partition against 0.931 for the 2 dB construction is a small difference, and it should not be over-read. The FAI-to-label step is calibrated to reproduce the crisp reference, so calibration absorbs much of whatever the partition changes. **Agreement with the WHO grade is therefore a weak test of the partition's value.**

This is consistent with the manuscript's own framing, that its contribution is "gradation and interpretability, not higher agreement with the reference". The evidence for the partition has to be the borderline behaviour and the graded outputs, which is where the reviewer's demand for a wider transition band and per-class reporting actually bites.

8. Corrections required in the manuscript

Parameter	Currently stated	Corrected
Weighted kappa	0.95	0.946
Overall agreement	97.1%	96.5%
Borderline agreement	88.0%	85.1%
Clear-case agreement	99.2%	99.1%
MAE	4.5 dB	4.35 dB
Spearman ρ	0.82	0.812

The borderline figure is the material one, being a headline number in the abstract and the quantity the review specifically probed. The rest are within rounding.

Also to add: the per-class table above, which answers the reviewer's request that the majority-class inflation in the aggregate metrics be made visible.

9. What this changes about the paper's position

The reproducibility failure is closed. The repository now contains the Ruspini implementation, the pipeline documenting it, the tests, and the verification scripts, and the numbers the manuscript reports come out of that pipeline.

Two things remain genuinely open:

The comparator baseline. The reviewer is right that regressing PTA-4 from seven frequency thresholds, four of which are the arithmetic terms of the target, is a

reconstruction task rather than a clinical benchmark. kappa = 0.981 for XGBoost demonstrates it. The reviewer's proposed alternative is a probabilistic multi-class model — ordinal logistic regression or multi-class gradient boosting — benchmarked against the fuzzy membership vectors with Brier scores and cross-entropy. That comparison does not currently exist and is the largest remaining methodological gap.

The membership overlap, now measured rather than argued. The partition supplies transition bands of 10 to 13.6 dB in place of 2 dB, which is what the reviewer asked for, but the consequence is a lower borderline agreement with the crisp reference by construction. That needs stating as an expected effect rather than presenting the wider band as a free improvement.

10. Artefacts

Code `fuzzy_audiogram/ruspini.py` — the construction, `verify`, `core_repairs`
`scripts/pipeline_participant.py` — two-stage `optimize_mfs`, fixed calibration
`tests/test_ruspini.py` — 16 unit tests `tests/test_pipeline_ruspini.py` — end-to-end check
`tests/diagnose_calibration.py` — reproduces the guard failure `tests/reoptimise_labels.py` — labelling schemes from saved predictions `tests/calibrate_train_eval_test.py` — train-fit, test-score comparison

Data `/home/hermes/.hermes/home/nhanes_raw/` — the six CDC files, 25 MB `data/output_participant/` — fresh run outputs `archive/participant_run_2db_overlap/` — the previous run, kept for comparison

Commits `3772e70` — Ruspini partition implementation `10c42fa` — calibration fix

11. Correction to an earlier claim

At one point in this work I reported a bug at line 229 of `pipeline_participant.py` — a comprehension written `for r in PTA_IDX` rather than `for i in PTA_IDX`, which raises `TypeError`. That was a misreading. The file reads `for i in PTA_IDX` and always has; git confirms it is unchanged since the commit. A reproduction that appeared to confirm the bug was testing my own mistyping rather than the file. There is no defect at that line.